DATE: DAY:

→ example 1

Simple calculator

Input

2+3*4

Dry run

1. Lexer converts input into tokens

   2 , + , 3 , * , 4

2. Parser checks operator precedence * is solved first

3. 3*4 = 12

4 2+12 = 14 → output

→ example 2 : calculator with variables

Input

a = 5

b = 3

a+b*2

c = a+1

Dry run

→ input a = 5

variable	value
a	5

→ input b = 3

variable	value
a	5
b	3

→ input = a+b*2

→ b×2 = 3*2 = 6

add

a+6 = 5+6 = 11

output = 11

DATE:...........

DAY:...........

input c = a+1

a = 5

5+1 = 6

variable c = 6

variable	Value
a	5
b	3
c	6

step 5

input c

value of c = 6

output = 6

example 3 : Boolean expression

Input 1

→ true and false

Dry run

true = 1

false = 0

1 and 0 = 0

Output = false

Input 2

not (true or false)

Dry run

1 → true or false

→ 1 or 0 = 1

2 → Apply Not

not 1 = 0

output = false

Input 3

Dry run

1. solve bracket

false or true

• 0 or 1 = 1

2. true and 1

• 1 and 1 = 1

output = true